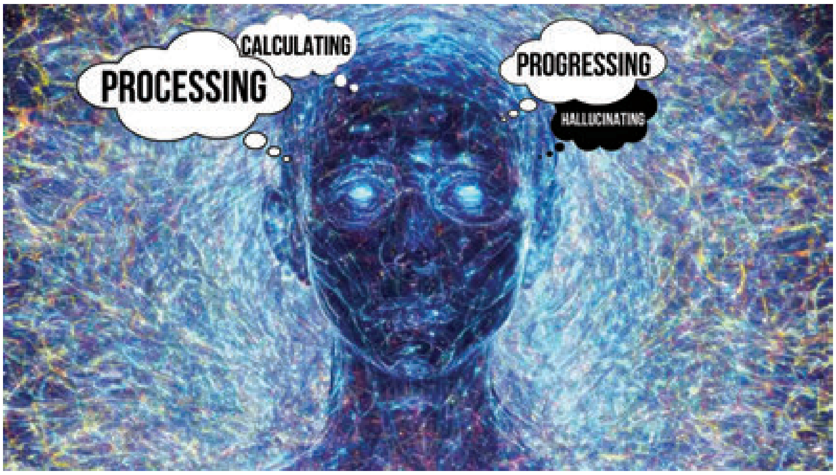

AI hallucination in LLM and beyond

Will it ever be fixed?

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Despite going mainstream two years ago, Generative AI products and services are arguably still in their infancy, and you just can't stop marvelling at their potent, transformative power. Even this early in its adoption curve, GenAI continues to impress. With broad consensus on GenAI as the next best thing since sliced bread, capable of responding to our whims and fancies better than our own wildest imagination, the honeymoon period is well and truly on. It seems these AI chatbots or text-to-image generators can do no wrong. Unless, of course, they do – at which point the honeymoon ends rather abruptly.

Just like us mere mortals, GenAI isn't without its flaws. Sometimes subtle, sometimes glaringly obvious. In its myriad attempts to conjure up text and images out of thin air, AI can have a tendency to make factual mistakes. In other words, hallucinate. These are instances where GenAI models produce incorrect, illogical or purely nonsensical output amounting to beautifully wrapped gibberish.

From Google Gemini's historically inaccurate images to Meta AI's gender biased pictures, whether it's ChatGPT's imaginary academic citations for generative text or Microsoft Edge's Bing Copilot giving erroneous information, these mistakes are noteworthy. Call it inference failure or Woke AI, they're all shades of AI hallucinations on display. Needless to say these AI hallucinations have been shocking, embarrassing and deeply concerning, giving even the most ardent of GenAI evangelists and gung-ho AI fans some serious pause. In fact, take any LLM (one of the pillars of GenAI currently) out there, it's guaranteed to make mistakes in something as simple as document summarisation. No jokes!

Researchers have created a public leaderboard on GitHub to track the hallucination rates in popular LLMs. They built an AI model to detect hallucinations in LLM outputs, feeding 1000 short documents to various AI models and measuring the rate of factual consistency and hallucination in their output. The models were also measured by their answer rate and average summary length. According to their leaderboard, some of the LLMs with the lowest hallucination rates are GPT-4 Turbo, Snowflake Arctic, and Intel Neural Chat 7B. They're also in the process of building a leaderboard on citation accuracy of LLMs – a crucial hurdle to overcome in terms of improving factual consistency.

Why does AI hallucinate?

AI hallucinations in popular LLMs like Llama 2 (70 billion parameters), GPT-3.5 (175 billion parameters), Claude Sonnet (70 billion parameters), etc, are all ultimately linked to their training data. Despite its gigantic size, if the training data of these LLMs had built-in bias of some kind, the generative AI output of these LLMs can have hallucinated facts that try to reinforce and transfer that bias in some form or another – similar to the Google Gemini blunders, for example. On the other end of the spectrum, absence of enough variety of data on any given subject can also lead to AI hallucinations every time the LLM is prompted on a topic it isn't well-versed to answer with authority.

If an LLM is trained on a mix of code and natural language-based data, it's very likely to hallucinate nonsensical code if it encounters a programming concept outside its training dataset. If the initial training data of image generation models like Midjourney or Stable Diffusion, which were trained on hundreds of billions of parameters, had a majority of images of Western architecture, for instance, their output will struggle to generate realistic or believable images of traditional Indian architecture, leading their models to invent or hallucinate a mish-mash of architectural variations that don't pass muster.